Chapter 21

Equivalent Representations, Sums, and Products

Road map

The same abstract representation may be written on different vector spaces or in different bases. We first make this statement precise through *equivalence of representations* and simultaneous similarity of all representing matrices. We then prove, by averaging an inner product over a finite group, that every finite-group representation on a finite-dimensional complex inner-product space is equivalent to a unitary representation. The last two parts construct new representations from old ones: the direct sum acts independently on two summands, whereas the tensor product acts simultaneously on two factors.

21.1 Equivalent representations

Let G be a group, let V, V' be vector spaces over the same field $\mathbb{K}$, and let

$$D : G \longrightarrow GL(V), \qquad D' : G \longrightarrow GL(V')$$

be representations.

Definition 21.1: Equivalent representations

The representations D and D' are *equivalent*, written $D \sim D'$, if there exists a vector-space isomorphism $S : V \to V'$ such that, for every $a \in G$,

$$D(a) = S^{-1} D'(a) S. \tag{21.1}$$

Equivalently,

$$D'(a)S = SD(a) \qquad (a \in G).$$

The isomorphism S is called an *intertwining isomorphism*, or simply an *intertwiner*.

The last equation says exactly that the following diagram commutes:

$$
\begin{tikzcd}
V \arrow[r, "D(a)"] \arrow[d, "S"'] & V \arrow[d, "S"] \\
V' \arrow[r, "D'(a)"'] & V'
\end{tikzcd}
$$

图 **21.1:** Equivalence means that applying $D(a)$ and then changing vector space gives the same answer as changing vector space first and applying $D'(a)$.

Indeed, starting with $v \in V$, the upper-right route produces

$$S(D(a)v),$$

whereas the lower-left route produces

$$D'(a)(Sv).$$

Equality for every v is precisely $SD(a) = D'(a)S$. Thus equivalent representations describe the same group action after the vectors have merely been relabelled by S.

21.1.1 Why this is an equivalence relation

Equivalence of representations satisfies the three axioms of an equivalence relation.

1. **Reflexivity.** Taking $S = \mathrm{id}_V$ gives $D(a) = S^{-1}D(a)S$, so $D \sim D$.
2. **Symmetry.** If $D(a) = S^{-1}D'(a)S$, then multiplication by S on the left and S^{-1} on the right gives
$$D'(a) = (S^{-1})^{-1}D(a)S^{-1}.$$
Hence $D' \sim D$, with intertwiner S^{-1}.
3. **Transitivity.** Suppose
$$D(a) = S^{-1}D'(a)S, \qquad D'(a) = T^{-1}D''(a)T.$$
Substitution yields
$$D(a) = S^{-1}T^{-1}D''(a)TS = (TS)^{-1}D''(a)(TS).$$
Thus $D \sim D''$, with intertwiner TS.

It is therefore natural to classify representations only up to equivalence.

If $V = V'$, Equation (21.1) says that the operators $D(a)$ and $D'(a)$ are similar. One must not overlook an important strengthening: there must be *one fixed* invertible map S that works simultaneously for every $a \in G$. Allowing a different similarity transformation for each group element would not define equivalence of representations.

21.2 Matrix representations and changes of basis

Assume that V is n-dimensional. Choose an ordered basis

$$\mathcal{B} = (e_1, \ldots, e_n)$$

and let

$$T_{\mathcal{B}} : V \longrightarrow \mathbb{K}^n, \qquad v = \sum_{i=1}^{n} v_i e_i \longmapsto [v]_{\mathcal{B}} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix},$$

be the coordinate isomorphism. The matrix representation determined by $\mathcal{B}$ is

$$D_{\mathcal{B}}(a) = T_{\mathcal{B}} D(a) T_{\mathcal{B}}^{-1}. \tag{21.2}$$

Choose a second ordered basis

$$\mathcal{B}' = (e'_1, \ldots, e'_n)$$

with coordinate isomorphism $T_{\mathcal{B}'}$, and define the coordinate change

$$S := T_{\mathcal{B}'} T_{\mathcal{B}}^{-1} : \mathbb{K}^n \longrightarrow \mathbb{K}^n. \tag{21.3}$$

Thus

$$[v]_{\mathcal{B}'} = S[v]_{\mathcal{B}}.$$

Using Equation (21.2) twice,

$$\begin{aligned} D_{\mathcal{B}'}(a) &= T_{\mathcal{B}'} D(a) T_{\mathcal{B}'}^{-1} \\ &= \left(T_{\mathcal{B}'} T_{\mathcal{B}}^{-1}\right)\left(T_{\mathcal{B}} D(a) T_{\mathcal{B}}^{-1}\right)\left(T_{\mathcal{B}} T_{\mathcal{B}'}^{-1}\right) \\ &= S D_{\mathcal{B}}(a) S^{-1}. \end{aligned}$$

Equivalently,

$$D_{\mathcal{B}}(a) = S^{-1} D_{\mathcal{B}'}(a) S. \tag{21.4}$$

This is exactly the equivalence relation from the previous section.

$$\begin{tikzcd}
V \arrow[r, "D(a)"] \arrow[d, "T_{\mathcal{B}}"'] & V \arrow[d, "T_{\mathcal{B}}"] \\
\mathbb{K}^n_{\mathcal{B}} \arrow[r, "D_{\mathcal{B}}(a)"] \arrow[d, "S"'] & \mathbb{K}^n_{\mathcal{B}} \arrow[d, "S"] \\
\mathbb{K}^n_{\mathcal{B}'} \arrow[r, "D_{\mathcal{B}'}(a)"'] & \mathbb{K}^n_{\mathcal{B}'}
\end{tikzcd}$$

图 **21.2:** The same abstract operator $D(a)$, expressed in two coordinate systems. Every rectangle commutes.

Convention check

Some books define the change-of-basis matrix in the opposite direction, $P = T_{\mathcal{B}} T_{\mathcal{B}'}^{-1} = S^{-1}$. They consequently write $D_{\mathcal{B}'}(a) = P^{-1} D_{\mathcal{B}}(a) P$. The formulas look reversed, but the mathematical statement is identical once the direction of the coordinate map has been fixed.

21.3 Unitarising a finite-group representation

Let V be a finite-dimensional complex inner-product space with original inner product $\langle v, w\rangle$. Recall that the inner product is conjugate linear in its first argument and linear in its second argument.

Theorem 21.1: Every finite-group representation is unitarisable

Let G be a finite group and let

$$D : G \longrightarrow GL(V)$$

be a representation on a finite-dimensional complex inner-product space $(V, \langle\cdot,\cdot\rangle)$. Then D is equivalent to a unitary representation.

证明. The given operators $D(a)$ need not preserve the original inner product. We repair this by averaging all of its translates over the group. Define

$$\{v, w\} := \frac{1}{|G|} \sum_{a\in G} \langle D(a)v, D(a)w\rangle\,. \tag{21.5}$$

The factor $1/|G|$ is a convenient normalization: it does not affect positivity or invariance, but makes the operation a literal group average.

We verify carefully that $\{\cdot,\cdot\}$ is an inner product.

1. **Sesquilinearity.** Each summand is conjugate linear in v and linear in w, because both $D(a)$ and the original inner product have these properties. A finite sum and a positive scalar multiple preserve them.

2. **Conjugate symmetry.**

$$\{w, v\} = \frac{1}{|G|} \sum_{a\in G} \langle D(a)w, D(a)v\rangle = \overline{\{v, w\}}.$$

3. **Positive definiteness.** For every v,

$$\{v, v\} = \frac{1}{|G|} \sum_{a\in G} \|D(a)v\|^2 \geq 0.$$

If $v \neq 0$, then the term corresponding to the identity e is $\|D(e)v\|^2 = \|v\|^2 > 0$. Therefore $\{v, v\} > 0$.

The crucial property is invariance under D. Fix $b \in G$. Then

$$\begin{aligned}\{D(b)v, D(b)w\} &= \frac{1}{|G|}\sum_{a\in G}\langle D(a)D(b)v, D(a)D(b)w\rangle \\ &= \frac{1}{|G|}\sum_{a\in G}\langle D(ab)v, D(ab)w\rangle\,.\end{aligned}$$

Right multiplication by b,

$$G \longrightarrow G, \qquad a \longmapsto ab,$$

is a bijection, with inverse $c \mapsto cb^{-1}$. Consequently the elements ab, as a runs through G, run through every element of G exactly once. Relabelling $c = ab$ therefore gives

$$\{D(b)v, D(b)w\} = \frac{1}{|G|}\sum_{c\in G}\langle D(c)v, D(c)w\rangle = \{v, w\}.$$

Thus D is unitary with respect to the averaged inner product.

It remains to express this as an equivalent representation that is unitary with respect to the *original* inner product. Choose an orthonormal basis $(e_1, \ldots, e_n)$ for $\langle\cdot,\cdot\rangle$, and choose an orthonormal basis $(q_1, \ldots, q_n)$ for $\{\cdot,\cdot\}$. Define the linear isomorphism $S : V \to V$ by

$$Se_i = q_i \qquad (1 \le i \le n).$$

For

$$v = \sum_i v_i e_i, \qquad w = \sum_j w_j e_j,$$

we compute

$$\begin{aligned}\{Sv, Sw\} &= \left\{\sum_i v_i q_i, \sum_j w_j q_j\right\} \\ &= \sum_{i,j} v_i^* w_j \{q_i, q_j\} \\ &= \sum_{i,j} v_i^* w_j \delta_{ij} = \sum_i v_i^* w_i = \langle v, w\rangle\,.\end{aligned}$$

Equivalently,

$$\{x, y\} = \left\langle S^{-1}x, S^{-1}y\right\rangle \qquad (x, y \in V). \tag{21.6}$$

Now define

$$D'(a) := S^{-1}D(a)S.$$

By construction $D' \sim D$. For $v, w \in V$, use Equation (21.6) and the invariance just proved:

$$\begin{aligned}\left\langle D'(a)v, D'(a)w\right\rangle &= \left\langle S^{-1}D(a)Sv, S^{-1}D(a)Sw\right\rangle \\ &= \{D(a)Sv, D(a)Sw\} \\ &= \{Sv, Sw\} \\ &= \langle v, w\rangle\,.\end{aligned}$$

Thus D' is unitary in the original inner product, as required. ∎

Infinite dimension and compact groups

The argument also applies when $\dim V = \infty$, provided the analytic conditions making the averaged inner product and the resulting isomorphism meaningful are satisfied. Its essential ingredient is an invariant, normalized group average.
For a finite group the average is

$$M(f) := \frac{1}{|G|} \sum_{a \in G} f(a)$$

and, for a compact Lie group such as $SO(n)$, it is replaced by integration against normalized Haar measure μ:

$$M(f) := \int_G f(a)\, \mathrm{d}\mu(a), \qquad \mu(G) = 1.$$

The Haar-invariance identity

$$\int_G f(ab)\, \mathrm{d}\mu(a) = \int_G f(a)\, \mathrm{d}\mu(a),$$

replaces the relabelling step in the finite-group proof. Under the usual continuity and finite-dimensionality assumptions, every representation of a compact group can therefore be made unitary.

21.4 Direct sums of vector spaces and linear maps

21.4.1 Sums of subspaces

A subset $U \subseteq V$ is a *linear subspace* if

$$u_1, u_2 \in U, \quad \alpha_1, \alpha_2 \in \mathbb{K} \quad \Longrightarrow \quad \alpha_1 u_1 + \alpha_2 u_2 \in U.$$

For two subspaces $U_1, U_2 \leq V$, their sum is

$$U_1 + U_2 := \{u_1 + u_2 : u_1 \in U_1,\ u_2 \in U_2\}. \tag{21.7}$$

This is again a subspace. In finite dimension,

$$\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2). \tag{21.8}$$

The subtraction is necessary because the directions lying in the intersection were counted once in $\dim U_1$ and once again in $\dim U_2$.

Definition 21.2: Internal direct sum

The sum $U_1 + U_2$ is a *direct sum*, denoted $U_1 \oplus U_2$, if $U_1 \cap U_2 = \{0\}$. Equivalently, in finite dimension,

$$\dim(U_1 + U_2) = \dim U_1 + \dim U_2.$$

Proposition 21.1: Direct sum and unique decomposition

The sum $U_1 + U_2$ is direct if and only if every $v \in U_1 + U_2$ has a unique expression

$$v = u_1 + u_2, \qquad u_1 \in U_1, \quad u_2 \in U_2.$$

证明. First suppose $U_1 \cap U_2 = \{0\}$, and suppose

$$v = u_1 + u_2 = u_1' + u_2'.$$

Rearranging gives

$$u_1 - u_1' = u_2' - u_2.$$

The left-hand side lies in U_1, while the right-hand side lies in U_2. Their common value is therefore in $U_1 \cap U_2 = \{0\}$. Consequently $u_1 = u_1'$ and $u_2 = u_2'$, proving uniqueness.

Conversely, suppose every decomposition is unique. If $x \in U_1 \cap U_2$, then

$$x = x + 0 = 0 + x$$

are two decompositions with components in U_1 and U_2. Uniqueness forces $x = 0$. Hence $U_1 \cap U_2 = \{0\}$. ■

21.4.2 Direct sums of maps

Let $f_1 : U_1 \to U_1$ and $f_2 : U_2 \to U_2$ be linear. On $U_1 \oplus U_2$, define

$$(f_1 \oplus f_2)(u_1 + u_2) := f_1(u_1) + f_2(u_2). \tag{21.9}$$

The unique-decomposition property makes this definition unambiguous. Linearity follows component by component. For example,

$$\begin{aligned}(f_1 \oplus f_2)\big(\alpha(u_1 + u_2) + \beta(u_1' + u_2')\big) &= f_1(\alpha u_1 + \beta u_1') + f_2(\alpha u_2 + \beta u_2') \\ &= \alpha(f_1 \oplus f_2)(u_1 + u_2) + \beta(f_1 \oplus f_2)(u_1' + u_2').\end{aligned}$$

Choose a basis

$$\mathcal{B}_1 = (u_1^{(1)}, \dots, u_{n_1}^{(1)}) \quad \text{of } U_1$$

and a basis

$$\mathcal{B}_2 = (u_1^{(2)}, \dots, u_{n_2}^{(2)}) \quad \text{of } U_2.$$

Their concatenation $\mathcal{B}_1 \,\|\, \mathcal{B}_2$ is a basis of $U_1 \oplus U_2$. If $A_1 = [f_1]_{\mathcal{B}_1}$ and $A_2 = [f_2]_{\mathcal{B}_2}$, then

$$[f_1 \oplus f_2]_{\mathcal{B}_1 \| \mathcal{B}_2} = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}. \tag{21.10}$$

The zero off-diagonal blocks record that $f_1 \oplus f_2$ never mixes a vector from one summand into the other.

21.5 Direct sums of representations

Definition 21.3: Direct-sum representation

Let

$$D_1 : G \longrightarrow GL(V_1), \qquad D_2 : G \longrightarrow GL(V_2)$$

be representations. Their *direct sum* is the map

$$D_1 \oplus D_2 : G \longrightarrow GL(V_1 \oplus V_2)$$

defined by

$$(D_1 \oplus D_2)(a)(v_1 + v_2) = D_1(a)v_1 + D_2(a)v_2. \tag{21.11}$$

In operator notation,

$$(D_1 \oplus D_2)(a) = D_1(a) \oplus D_2(a).$$

Let us verify that this is genuinely a representation. For $a, b \in G$ and $v = v_1 + v_2$,

$$\begin{aligned}(D_1 \oplus D_2)(a)(D_1 \oplus D_2)(b)v &= (D_1 \oplus D_2)(a)\big(D_1(b)v_1 + D_2(b)v_2\big)\\ &= D_1(a)D_1(b)v_1 + D_2(a)D_2(b)v_2\\ &= D_1(ab)v_1 + D_2(ab)v_2\\ &= (D_1 \oplus D_2)(ab)v.\end{aligned}$$

Moreover,

$$(D_1 \oplus D_2)(e) = I_{V_1} \oplus I_{V_2} = I_{V_1 \oplus V_2}.$$

In bases adapted to $V_1 \oplus V_2$, the representing matrices are

$$[D_1 \oplus D_2](a) = \begin{pmatrix} D_1(a) & 0 \\ 0 & D_2(a) \end{pmatrix}, \qquad \dim(D_1 \oplus D_2) = \dim D_1 + \dim D_2. \tag{21.12}$$

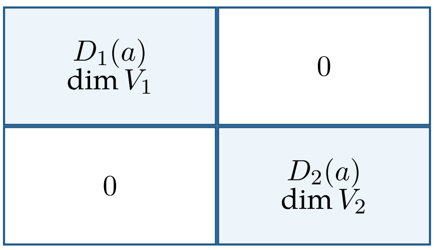

图 **21.3:** The block-diagonal matrix of a direct-sum representation in a basis adapted to the two summands.

The block-diagonal appearance is basis-dependent. An arbitrary change of basis on $V_1 \oplus V_2$ can mix the two displayed blocks and make the matrix look dense. The invariant statement is that the representation space contains two complementary invariant subspaces. In a later chapter this observation will lead to reducible and irreducible representations.

21.6 Tensor products of vector spaces and maps

21.6.1 The tensor-product space

Let V_1, V_2 be vector spaces over $\mathbb{K}$. Their tensor product is the vector space

$$V_1 \otimes V_2 = \operatorname{span}\{v_1 \otimes v_2 : v_1 \in V_1,\ v_2 \in V_2\},$$

subject to the bilinearity relations

$$(v_1 + v_1') \otimes v_2 = v_1 \otimes v_2 + v_1' \otimes v_2, \tag{21.13}$$
$$v_1 \otimes (v_2 + v_2') = v_1 \otimes v_2 + v_1 \otimes v_2', \tag{21.14}$$
$$\lambda(v_1 \otimes v_2) = (\lambda v_1) \otimes v_2 = v_1 \otimes (\lambda v_2), \qquad \lambda \in \mathbb{K}. \tag{21.15}$$

In particular,

$$0 \otimes v_2 = (0+0) \otimes v_2 = 0 \otimes v_2 + 0 \otimes v_2,$$

so subtracting $0 \otimes v_2$ from both sides gives

$$0 \otimes v_2 = 0.$$

Similarly $v_1 \otimes 0 = 0$.

A vector of the form $v_1 \otimes v_2$ is called a *simple* or *pure* tensor. A general tensor is a finite sum of pure tensors and need not itself be pure.

Suppose

$$(v_1^{(1)}, \ldots, v_{n_1}^{(1)}) \quad \text{and} \quad (v_1^{(2)}, \ldots, v_{n_2}^{(2)})$$

are bases of V_1 and V_2. Then

$$\{v_i^{(1)} \otimes v_j^{(2)} : 1 \le i \le n_1,\ 1 \le j \le n_2\} \tag{21.16}$$

is a basis of $V_1 \otimes V_2$. There are $n_1 n_2$ basis elements, and therefore

$$\dim(V_1 \otimes V_2) = (\dim V_1)(\dim V_2). \tag{21.17}$$

To see explicitly why the spanning property holds, write

$$v_1 = \sum_i x_i v_i^{(1)}, \qquad v_2 = \sum_j y_j v_j^{(2)}.$$

Bilinearity gives

$$v_1 \otimes v_2 = \sum_{i,j} x_i y_j \big(v_i^{(1)} \otimes v_j^{(2)}\big).$$

Since arbitrary tensors are sums of pure tensors, the displayed tensors span the whole tensor product.

21.6.2 Tensor products of linear maps

For linear maps $f_1 : V_1 \to V_1$ and $f_2 : V_2 \to V_2$, define

$$f_1 \otimes f_2 : V_1 \otimes V_2 \longrightarrow V_1 \otimes V_2$$

first on pure tensors by

$$(f_1 \otimes f_2)(v_1 \otimes v_2) := f_1(v_1) \otimes f_2(v_2), \tag{21.18}$$

and then extend linearly to all tensors. The bilinearity relations ensure that the definition is well-defined.

Tensor products of maps obey the useful composition rule

$$(f_1 \otimes f_2)(g_1 \otimes g_2) = (f_1 g_1) \otimes (f_2 g_2). \tag{21.19}$$

It is enough to check this on a pure tensor:

$$\begin{aligned}(f_1 \otimes f_2)(g_1 \otimes g_2)(v_1 \otimes v_2) &= (f_1 \otimes f_2)\big(g_1(v_1) \otimes g_2(v_2)\big) \\ &= f_1(g_1(v_1)) \otimes f_2(g_2(v_2)) \\ &= \big((f_1 g_1) \otimes (f_2 g_2)\big)(v_1 \otimes v_2).\end{aligned}$$

21.7 Tensor products of representations

Definition 21.4: Tensor-product representation

Let

$$D_1 : G \longrightarrow GL(V_1), \qquad D_2 : G \longrightarrow GL(V_2)$$

be representations. Their *tensor product*, also called the *product representation*, is

$$D_1 \otimes D_2 : G \longrightarrow GL(V_1 \otimes V_2)$$

defined by

$$(D_1 \otimes D_2)(a) := D_1(a) \otimes D_2(a). \tag{21.20}$$

Thus, on a pure tensor,

$$(D_1 \otimes D_2)(a)(v_1 \otimes v_2) = D_1(a)v_1 \otimes D_2(a)v_2.$$

The representation law follows directly from Equation (21.19):

$$\begin{aligned}(D_1 \otimes D_2)(a)(D_1 \otimes D_2)(b) &= \big(D_1(a)D_1(b)\big) \otimes \big(D_2(a)D_2(b)\big) \\ &= D_1(ab) \otimes D_2(ab) \\ &= (D_1 \otimes D_2)(ab).\end{aligned}$$

Also,

$$(D_1 \otimes D_2)(e) = I_{V_1} \otimes I_{V_2} = I_{V_1 \otimes V_2}.$$

Therefore the definition indeed produces a representation, of dimension

$$\dim(D_1 \otimes D_2) = (\dim D_1)(\dim D_2). \tag{21.21}$$

In the product basis from Equation (21.16), the matrix of $(D_1 \otimes D_2)(a)$ is the Kronecker product of the two matrices. If

$$D_1(a) = (A_{ki}), \qquad D_2(a) = (B_{\ell j}),$$

then

$$\begin{aligned}(D_1(a) \otimes D_2(a))\big(v_i^{(1)} \otimes v_j^{(2)}\big) &= \left(\sum_k A_{ki} v_k^{(1)}\right) \otimes \left(\sum_\ell B_{\ell j} v_\ell^{(2)}\right) \\ &= \sum_{k,\ell} A_{ki} B_{\ell j} \big(v_k^{(1)} \otimes v_\ell^{(2)}\big).\end{aligned}$$

Hence the matrix entry with row pair (k, ℓ) and column pair (i, j) is

$$(D_1(a) \otimes D_2(a))_{(k,\ell),(i,j)} = A_{ki} B_{\ell j}.$$

Direct sum versus tensor product

The two constructions combine dimensions in different ways:

$$\dim(D_1 \oplus D_2) = \dim D_1 + \dim D_2, \qquad \dim(D_1 \otimes D_2) = (\dim D_1)(\dim D_2).$$

In a direct sum, a group element acts on either component separately. In a tensor product, the same group element acts on both factors at once.

What to retain from this chapter

1. Representations D and D' are equivalent if a single isomorphism S intertwines every group action:
$$D(a) = S^{-1}D'(a)S \qquad (a \in G).$$

2. Choosing a different basis gives an equivalent matrix representation. All representing matrices undergo the same similarity transformation.

3. For a finite group, averaging any inner product,
$$\{v, w\} = \frac{1}{|G|}\sum_{a\in G} \langle D(a)v, D(a)w\rangle\,,$$
produces a positive-definite G-invariant inner product. It follows that every finite-dimensional representation of a finite group is equivalent to a unitary representation.

4. The condition $U_1 \cap U_2 = \{0\}$ is equivalent to the uniqueness of every decomposition $u_1 + u_2$ in $U_1 \oplus U_2$.

5. Direct-sum representations act componentwise and have block-diagonal matrices in an adapted basis:
$$(D_1 \oplus D_2)(a) = \begin{pmatrix} D_1(a) & 0 \\ 0 & D_2(a) \end{pmatrix}.$$

6. Tensor products are bilinear, have the product basis $v_i^{(1)} \otimes v_j^{(2)}$, and satisfy
$$\dim(V_1 \otimes V_2) = \dim V_1\ \dim V_2.$$

7. The product representation acts by
$$(D_1 \otimes D_2)(a)(v_1 \otimes v_2) = D_1(a)v_1 \otimes D_2(a)v_2.$$